INVESTMENT COMMITTEE

Quantitative Equity Portfolio

Photonics & Technology Growth

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Quantitative Research | April 08, 2026

86.6%
Total Return

2.53
Sharpe Ratio

1.95×
Portfolio Beta

11
Positions Held

AGENDA

Presentation Structure

01 **Portfolio Overview**
Construction, tickers, sector themes

02 **Return Attribution**
Backtest vs SPY — Module 1

03 **Risk & VaR Analysis**
Three-method VaR, beta, ratios — Module 2

04 **Technical Signal Dashboard**
RSI, MA, Bollinger, Z-score — Module 3

05 **Limitations & Next Steps**
Honest constraints and v2 roadmap

METHODOLOGY STACK

Python · yfinance · pandas

Backtester → VaR → Signals

R Markdown → PDF Memo

GitHub · Colab · Posit Cloud

PORTFOLIO OVERVIEW

Eleven Positions Across Five Sector Themes

Photonics

LITE COHR

Semiconductor Infra

AVGO MRVL AMD

Large-Cap Tech

GOOG

Commodity Hedge

GLD

CONSTRUCTION

Entry Date	**Aug 8, 2025**
2nd Tranche	**Aug 26, 2025**
AMD Top-Up	**Sep 13, 2025**
Capital	**~$66.63 deployed**
Structure	**Fractional shares**
Rebalancing	**None — buy & hold**

PORTFOLIO OVERVIEW

Position Summary — Inception to April 2026

Ticker	Entry Price	Current Price	Return (%)	P&L ($)	Beta
LITE	$116.27	$815.75	601.6%	+$30.80	3.09
COHR	$115.44	$255.10	121.0%	+$7.79	3.02
VRT	$139.81	$262.30	87.6%	+$4.63	2.72
GOOG	$201.64	$303.93	50.7%	+$2.54	1.28
MRVL	$77.23	$109.38	41.6%	+$2.37	2.33
GLD	$308.55	$431.81	39.9%	+$2.01	0.53
EQIX	$762.69	$1,007.29	32.1%	+$1.51	0.44
AMD	$172.76	$221.53	28.2%	+$5.00	2.63
SNDR	$23.70	$27.44	15.8%	+$0.61	1.23
AVGO	$303.23	$333.97	10.1%	+$0.61	2.13
ANET	$139.18	$133.64	-4.0%	-$0.16	2.03

MODULE 1 · BACKTESTER

Portfolio vs SPY — August 2025 to April 2026

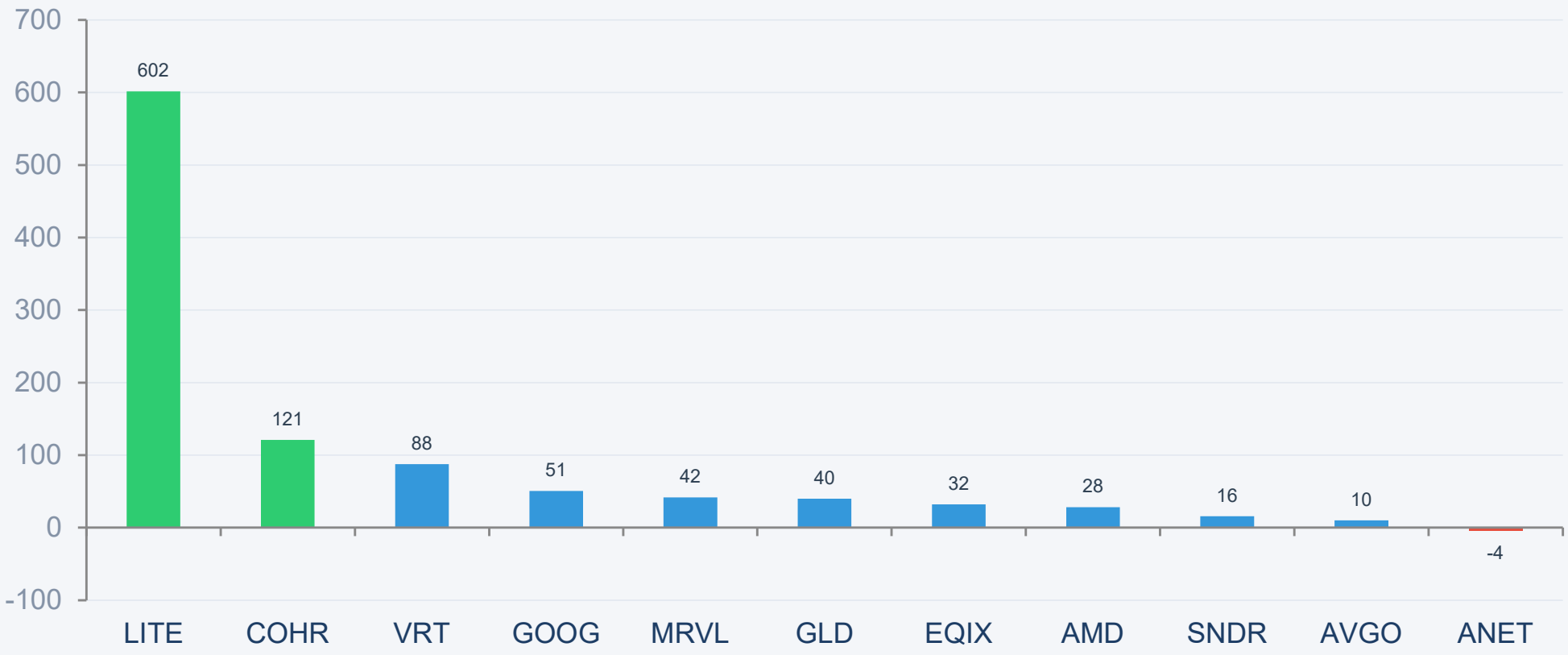
86.6%	~5%	81.6pp	165
Total Return	SPY Return	Outperformance	Trading Days

KEY OBSERVATIONS

- Portfolio value grew from $66.63 deployed to ~$124 — nearly doubling in 8 months
- LITE (601.6%) and COHR (121%) drove the majority of absolute returns
- Max drawdown of -10.4% observed Nov–Dec 2025; SPY drawdown was shallower
- DCA structure (Aug 8 + Aug 26 tranches) reduced timing risk on entry

MODULE 1 · BACKTESTER

Per-Ticker Return — Entry to Today



MODULE 2 · RISK & VAR

Value at Risk — Three Methodologies

Method	VaR 95%	VaR 99%	CVaR 95%	Note
Historical Simulation	-3.37%	-4.79%	-4.23%	PRIMARY — no distributional assumption
Parametric (Var-Cov)	-3.06%	-4.47%	-3.92%	Underestimates fat tails
Monte Carlo (10k sims)	-3.08%	-4.46%	-3.95%	Consistent with parametric

INTERPRETATION

Fat Tails Confirmed

Historical VaR (-3.37%) > Parametric (-3.06%). Actual loss distribution has heavier tails than normal.

Daily Loss Threshold

5% chance of losing more than $4.14 on any given trading day (Historical, 95% confidence).

Expected Shortfall

When the VaR threshold is breached, the average loss (CVaR) is $5.20 — 26% worse than the VaR alone.

MODULE 2 · RISK & VAR

Risk-Adjusted Performance Ratios

2.53

> 1.0 = good

Sharpe Ratio

Excess return per unit total risk

3.78

> Sharpe = favorable

Sortino Ratio

Penalizes downside vol only

0.426

Higher = better

Treynor Ratio

Excess return per unit market risk

73.4%

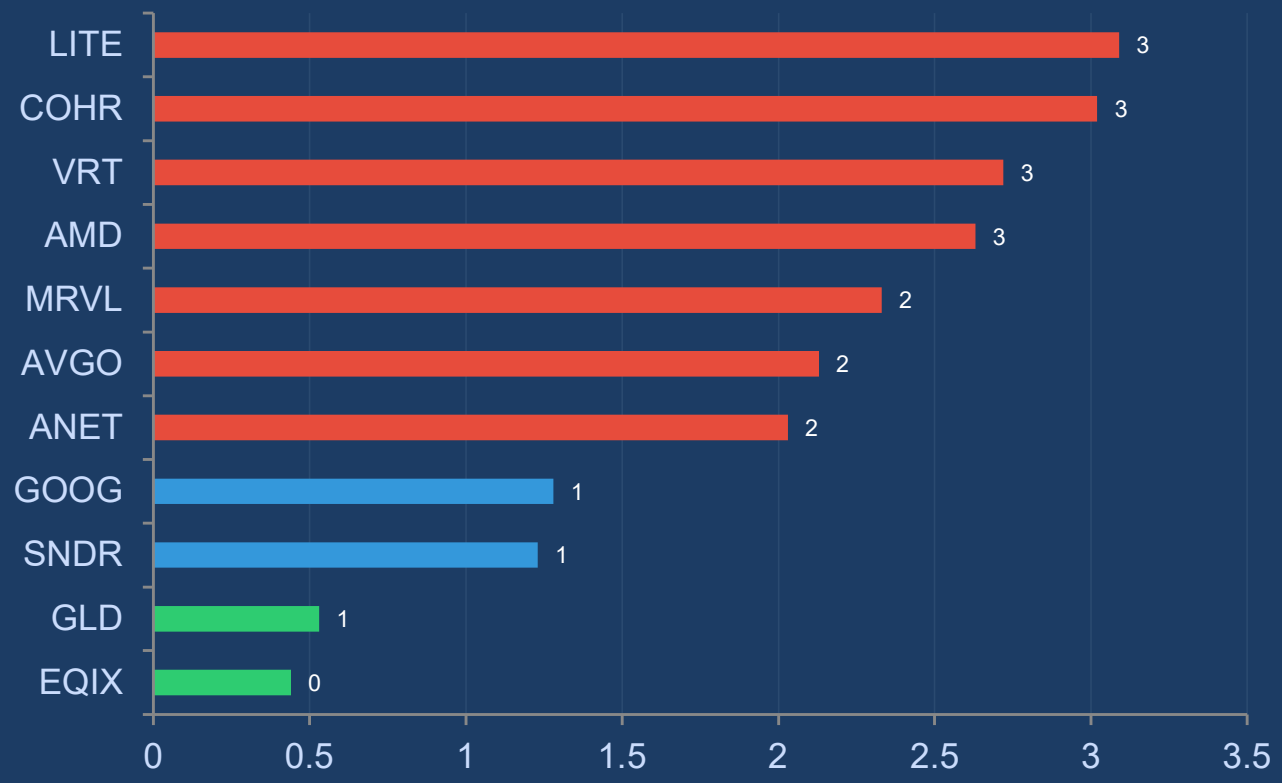
Positive = skill

Jensen's Alpha (Ann.)

Return above beta-explained benchmark

MODULE 2 · RISK & VAR

Per-Ticker Beta vs SPY — Concentration Risk



Low Beta < 1.0

EQIX, GLD — defensive anchors

Moderate 1.0–1.5

SNDR, GOOG — market-like moves

High Beta > 1.5

7 positions — amplified moves

KEY RISK

LITE + COHR both at beta ~3.0 and correlated at 0.72. A photonics correction would produce ~3× the SPY drawdown on these positions.

MODULE 2 · RISK & VAR

Correlation Analysis — Diversification Reality Check

NOTABLE CORRELATION PAIRS

Pair	Correlation	Level	Interpretation
LITE ↔ COHR	0.72	HIGH	Same photonics thesis — functionally one position
VRT ↔ LITE	0.53	MOD	Data center + photonics overlap in AI infra theme
VRT ↔ COHR	0.54	MOD	Same AI infra demand driver
AVGO ↔ LITE	0.50	MOD	Semiconductor supply chain linkage
EQIX ↔ SNDR	-0.00	LOW	True diversifiers — effectively uncorrelated
GLD ↔ all	~0.15	LOW	Gold acts as genuine portfolio hedge

Takeaway: Only one highly correlated pair (LITE↔COHR >0.70). Eleven positions overstate diversification — effective independent bets are closer to eight.

MODULE 3 · LIVE SIGNALS

Technical Signal Dashboard — April 08, 2026

Ticker	Price	RSI	RSI Signal	MA Signal	BB Signal	Z Signal	Score
COHR	$255.10	52.3	NEUTRAL	GOLDEN CROSS	MID BAND	NORMAL	+1
GLD	$431.81	47.3	NEUTRAL	GOLDEN CROSS	MID BAND	NORMAL	+1
GOOG	$303.93	54.8	NEUTRAL	GOLDEN CROSS	MID BAND	NORMAL	+1
VRT	$262.30	54.6	NEUTRAL	GOLDEN CROSS	MID BAND	NORMAL	+1
ANET	$133.64	53.3	NEUTRAL	GOLDEN CROSS	MID BAND	NORMAL	+1
AMD	$221.53	59.7	NEUTRAL	GOLDEN CROSS	NEAR UPPER	NORMAL	0
LITE	$815.75	59.9	NEUTRAL	GOLDEN CROSS	NEAR UPPER	NORMAL	0
EQIX	$1007.29	68.4	NEUTRAL	GOLDEN CROSS	NEAR UPPER	NORMAL	0
MRVL	$109.38	69.0	NEUTRAL	GOLDEN CROSS	NEAR UPPER	EXTENDED	-1
AVGO	$333.97	57.9	NEUTRAL	DEATH CROSS	NEAR UPPER	NORMAL	-2

No RSI extremes across all 11 positions — consistent with macro volatility compression (April 2026 tariff environment). Watch: AVGO death cross, MRVL z-score extended.

MODULE 3 · LIVE SIGNALS

Signal Definitions & Composite Scoring

RSI

14-period Wilder RSI

Oversold <30 = +1 | Neutral = 0 | Overbought >70 = -1

MA Cross

50-day vs 200-day SMA

Golden Cross (50>200) = +1 | Death Cross = -1

Bollinger

20-day, ±2σ bands

Near/Below Lower = +1 | Mid = 0 | Near/Above Upper = -1

Z-Score

60-day rolling mean & std

< -2σ (Cheap) = +1 | Normal = 0 | > +2σ (Extended) = -1

Volume

Ratio to 20-day avg volume

High Vol (>1.5×) = +1 | Normal = 0 | Low Vol (<0.5×) = -0.5

Composite Score = sum of all signal scores. Range: -4 (fully bearish) to +4 (fully bullish). Equal-weighted — a v2 limitation to address.

LIMITATIONS

Material Study Constraints — v1 Disclosure

Sample Size

165 trading days (~8 months). Sharpe, alpha, and all ratios carry wide confidence intervals. Not statistically robust.

Benchmark Selection

SPY returned only ~5% during the study period — an easy hurdle. QQQ and a photonics ETF benchmark comparison deferred to v2.

Concentration

LITE + COHR at 0.72 correlation = one photonics bet, not two. Eleven positions overstate diversification.

Static Beta

Beta calculated as a single point estimate. Rolling beta analysis would reveal significant regime variation — not yet implemented.

Hardcoded Portfolio NAV

VaR dollar estimates used a static $123 input rather than dynamic NAV. Percentage VaR is correct; dollar figures are approximate.

No Transaction Costs

Spreads, platform fees, and tax treatment excluded from all return and ratio calculations.

NEXT STEPS

Version 2 — Research Roadmap

01

Expand benchmark set

Add QQQ, RAYS (photonics ETF) as additional benchmarks for fairer comparison

02

Rolling beta analysis

Compute 30-day rolling beta to detect regime shifts around key market events

03

Dynamic NAV for VaR

Replace hardcoded portfolio value with live NAV calculation from Module 1 output

04

Confidence intervals

Bootstrap Sharpe and alpha estimates to report statistical uncertainty ranges

05

Transaction cost model

Incorporate spread estimates and tax lot accounting into net return calculations

06

12-month extension

Minimum 252 trading days (1 full year) before drawing definitive strategic conclusions

CONCLUSION & RECOMMENDATION

Five Observations for the Committee

1 **Performance merits monitoring —**

Sharpe 2.53, alpha 73.4% — strong risk-adjusted signal. Warrants continued observation and potential disciplined scale-up.

2 **AVGO — watch flag —**

Death cross + near upper band = distribution pattern. Underperforming portfolio peers. Reassess on next earnings catalyst.

3 **MRVL — limited near-term upside —**

Z-score extended (+2.79σ) + Bollinger near upper. Price appears stretched without a new fundamental catalyst.

4 **Photonics concentration is primary risk —**

LITE + COHR at beta ~3.0, corr 0.72. A sector correction produces ~3× SPY drawdown on these positions.

5 **12 months before conclusions —**

165 days is insufficient for statistical confidence. All ratios carry wide confidence intervals at current sample size.

Data: Yahoo Finance via yfinance | Python Modules 1–3 (Colab) | R Markdown Module 5 (Posit Cloud) | github.com/zaeemhasan/quant-portfolio-study